

Movie Ratings Forecast

Recommender Systems for Personalized Content Discovery

Netflix Prize Market Data 1999 – 2005

Aryan Sharma

Enrollment No. 24113024

Akhil Kumar

Enrollment No. 24117009

ABSTRACT

This report presents a comprehensive collaborative-filtering recommendation platform evaluated on the Netflix Prize dataset spanning over six years of user interactions. The system implements a modular pipeline comparing four models: a regularized user/item bias baseline, an item-based neighborhood KNN, Alternating Least Squares (ALS) matrix factorization, and Funk-SVD trained via mini-batch SGD.

While all models successfully outperform the baseline on rating prediction error, our findings reveal a stark divergence between accuracy metrics and ranking performance. The ALS model dominates ranking quality, delivering a 5.5× increase in MAP@10 (0.101) and a 78.6% user Hit-Rate, making it the optimal candidate for production-level personalized content discovery. Conversely, Funk-SVD yields the lowest overall RMSE (0.821) and the widest catalog coverage (64.3%), demonstrating the inherent trade-offs between precision and exploratory exposure in sparse, long-tailed interaction matrices.

Problem Overview

Streaming catalogues dwarf what any user can browse — value comes from surfacing the **right** titles for each person. We frame this as a **collaborative-filtering** problem with two complementary goals:

- **Rating prediction (regression)** — forecast the 1–5★ rating a user would give a movie. Scored by RMSE / MAE.
- **Top-N recommendation (ranking)** — produce an ordered list of unseen movies a user will enjoy. Scored by MAP, NDCG, Precision/Recall, Hit-Rate, Coverage.

The two goals diverge: accuracy is dominated by easy near-average ratings, while ranking is decided by the few items at the top of each list. We therefore optimize and judge **both** — and treat ranking as the primary business signal, since users experience the *list*, not the scores.

THE CORE OBSTACLE

Extreme sparsity.

Most user×movie cells are never observed — the filtered matrix is **~81% empty**, the raw Netflix matrix **~99%**.

The model must generalize from a thin set of interactions while staying robust to popularity bias and cold users. This is what makes low-rank **matrix factorization** the natural fit.

Approach



Data & split. Netflix Prize streamed in chunks, filtered to active users (≥ 50) and popular movies ($\geq 2,000$), capped to 40k users $\times$ 1,150 movies $\rightarrow$ 8.62M ratings, stored sparse (CSR). A **per-user 80/20 holdout** (seed 42) $\rightarrow$ **6.88M train / 1.74M test**, keeping every test user in train.

Evaluation. Identical split & code for every model. Each user is scored over all unseen movies; Top-10 (training items masked) is compared to held-out items rated ≥ 3.5 ★. Apples-to-apples throughout.

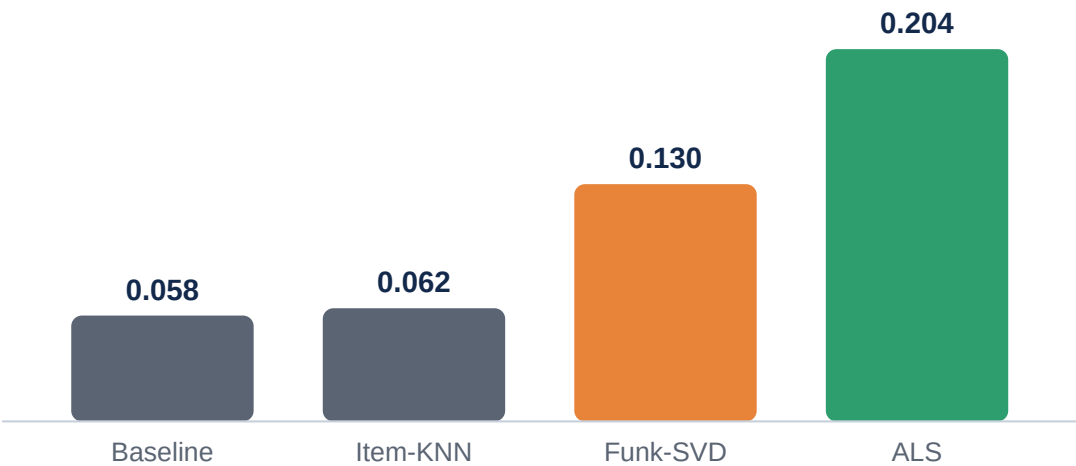
Model	Core idea
Baseline bias	$\mu + b_u + b_i$ — global mean plus regularized user/item bias. A strong near-trivial reference.
Item-KNN CF	Mean-centered cosine item similarity, top-k=200 neighbors over rating residuals.
ALS MF	Factorizes residuals via alternating ridge least-squares. 40 factors, reg=50.
Funk-SVD MF + SGD	Biased MF, jointly learns μ , biases & factors by SGD. 50 factors, 25 epochs.

All four share one BaseRecommender interface; the MF models build *on top of* the bias baseline — a “bias-first” design.

Experimental Results

Model	RMSE↓	MAE↓	MAP@10↑	NDCG@10↑	Prec@10↑	Hit@10↑	Cov@10↑	Fit (s)
Baseline (bias)	0.900	0.708	0.018	0.058	0.067	0.480	0.043	0.2
Item-KNN CF	0.843	0.657	0.024	0.062	0.064	0.404	0.602	2.6
ALS (MF)	0.829	0.642	0.101	0.204	0.175	0.786	0.392	67.8
Funk-SVD	0.821	0.641	0.055	0.130	0.130	0.681	0.643	226.7

NDCG@10 — ranking quality



Reading the results

- **ALS dominates ranking** — MAP@10 0.101 (5.5× baseline), NDCG 0.204, and a **78.6% hit-rate**: ~4 in 5 users get a relevant title in their Top-10.
- **Funk-SVD** wins pure rating error (RMSE 0.821) and **coverage (64%)**, but ranks below ALS — accuracy ≠ ranking.
- **Item-KNN** has decent RMSE but generic lists: 1,150 popular movies make neighborhoods blockbuster-heavy.
- **ALS is the efficiency sweet spot** — best ranking, trains 3× faster than Funk-SVD.

Key Insights

01

RMSE hides the story

The bias baseline scores RMSE 0.900 — because ratings skew high, predicting near the mean is hard to beat. Rating error alone is a poor recommender judge.

02

Ranking separates models

ALS lifts MAP@10 **5.5×** and NDCG@10 **3.5×** over baseline — gains that are invisible in RMSE. This validates the dual-metric design.

03

Accuracy ≠ good recs

Funk-SVD has the *lowest* RMSE yet ranks below ALS. Optimizing the wrong metric would have shipped the wrong model.

04

Factorization beats sparsity

Low-rank latent factors generalize across the 81–99% empty matrix far better than neighborhood similarity — the right inductive bias.

05

Bias-first design pays off

Removing user/item bias before factorizing residuals (ALS) yields the cleanest signal and the best ranking-per-second.

06

Coverage vs. relevance

Funk-SVD covers 64% of the catalogue, ALS 39%. Broader long-tail exposure can cost short-term precision — a real product trade-off.

Recommendation Examples

For two users we show titles they rated 5★ (the input signal) and the model's fresh recommendations with predicted ratings — none already seen.

User A
mainstream action / classic-cinema fan

RATED 5★ (input)	TOP RECS	pred ★
★ The Sixth Sense (1999)	1. Alien	4.55
★ Lethal Weapon (1987)	2. Aliens	4.46
★ The Maltese Falcon (1941)	3. Lost: Season 1	4.42
★ Batman Begins (2005)	4. Seven Samurai	4.35
★ Sixteen Candles (1984)	5. Hoop Dreams	4.27
	6. A Hard Day's Night	4.24

User B
cult / dark-comedy & prestige-TV fan

RATED 5★ (input)	TOP RECS	pred ★
★ Reservoir Dogs (1992)	1. Curb Your Enthusiasm S3	4.36
★ Fear and Loathing in LV (1998)	2. Lost: Season 1	4.32
★ Silence of the Lambs (1991)	3. The Simpsons S6	4.27
★ Sideways (2004)	4. Seven Samurai	4.27
★ Notorious (1946)	5. The Twilight Zone V36	4.24
	6. Family Guy Collection	4.19

Lists are **coherent with each user's taste** — A gets canonical action/sci-fi and acclaimed classics; B is steered to edgy comedy and prestige TV. The latent factors capture genuine taste structure, not just global popularity.

Recommendation

Deploy ALS as the production model

Best ranking metrics + fastest of the factor models — the strongest user-facing experience per second of compute.

Why ALS

- Top MAP@10 (0.101), NDCG@10 (0.204) and Hit-Rate (78.6%).
- Trains in ~68s — 3× faster than Funk-SVD via closed-form least squares.
- Bias-first residual factorization → clean, stable signal.

When to prefer Funk-SVD

- Lowest rating error (RMSE 0.821) — if score accuracy is the goal.
- Best catalogue coverage (64%) — to surface the long tail / aid discovery.
- Natural fit for an **ALS + Funk-SVD ensemble** with diversity re-ranking.

Next steps

Time-aware split

Implicit-feedback / BPR

Hyper-param search

Hybrid content features

Scale to full 17,770 titles

IN ONE LINE

Matrix factorization — and *how you measure it* — is what turns ratings into recommendations users actually feel.

ALS

recommended model

0.204

NDCG@10 (3.5× baseline)

78.6%

users helped (Hit@10)

8.62M

ratings · 4 models

Reproducibility: `data/preprocess.py` → `script/train.py` → `script/recommend.py` · seed 42 throughout

Thank you